Reverse Polish Notation

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The problem

- Evaluate postfix tokens with + - * / (integer division truncating toward zero).

Intuition

- Operands go on a stack; each operator pops two and pushes the result.

Approach

- 1. Push numbers
- 2. On an operator, pop right then left
- 3. Apply and push
- 4. Return the last value

Java solution

```java
public class ReversePolishNotation {
    public static int reversePolishNotation(String[] tokens) {
        ArrayDeque<Integer> st = new ArrayDeque<>();
        for (String t : tokens) {
            switch (t) {
                case "+" -> st.push(st.pop() + st.pop());
                case "*" -> st.push(st.pop() * st.pop());
                case "-" -> { int b = st.pop(); st.push(st.pop() - b); }
                case "/" -> { int b = st.pop(); st.push(st.pop() / b); }
                default -> st.push(Integer.parseInt(t));
            }
        }
        return st.pop();
    }
}
```

Complexity

- Time: O(n)
- Space: O(n)

Recap

- Operands go on a stack; each operator pops two and pushes the result.
- Complexity: O(n)